# Skateboarding as a Noisy Signal of Romantic Interest

A Valentine's Day Question

*A Bayesian decision model with risk preferences*

February 13, 2026

### Idea

Chris always skateboards, and Dumpling knows this. Dumpling may ask Chris to teach her. Asking *might* be a signal of romantic interest, but it can also reflect (i) genuine interest in learning to skate and/or (ii) a risk-seeking personality. This makes "asking" a *noisy* signal rather than a clean one.

### Risk and preferences

Skateboarding involves a small but non-zero probability of death $p \in (0, 1)$ and a disutility of death $D > 0$. Dumpling experiences the expected risk cost through her risk parameter $r > 0$:

$$\text{Perceived expected risk cost} = r\, p\, D.$$

We will allow risk preferences to vary across people.

### Private information (Dumpling's type)

Dumpling has two private traits:

- **Romantic-interest type** $t \in \{L, N\}$, where $L$ means she likes Chris romantically and $N$ means she is not romantically interested.
- **Risk type** $r \in \{r_A, r_S\}$ where $r_A > r_S > 0$. Interpret $r_A$ as risk-averse and $r_S$ as risk-seeking.

Let the priors be

$$\Pr(t = L) = \pi, \qquad \Pr(r = r_A) = \rho,$$

and assume (for simplicity) that $t$ and $r$ are independent.

### Action

Dumpling chooses

$$A \in \{0, 1\}, \qquad A = 1 \text{ means "ask Chris to teach me"}, \qquad A = 0 \text{ means "do not ask."}$$

**Benefits from asking**

If Dumpling asks, she may receive:

- a **bonding benefit** $B > 0$ if she is romantically interested ($t = L$),
- a **learning/hobby benefit** $G \geq 0$ from learning to skateboard (regardless of $t$).

**Decision rule (Dumpling)**

Dumpling compares the benefits to her perceived risk cost $rpD$.

If $t = L$:

$$U_N(A = 1 \mid L, r) = B + G - rpD, \qquad \Rightarrow \qquad A = 1 \iff B + G > rpD.$$

If $t = N$:

$$U_N(A = 1 \mid N, r) = G - rpD, \qquad \Rightarrow \qquad A = 1 \iff G > rpD.$$

**Two confounding cases (why the signal is messy)**

This framework cleanly captures the two situations that break the naive "asking means she likes you" story.

**(1) She likes you but is too risk-averse to ask.** This occurs when $t = L$ and $r = r_A$ but

$$B + G \leq r_A pD.$$

So she likes Chris, but the perceived cost of risk dominates.

**(2) She does not like you but is risk-seeking and asks anyway.** This occurs when $t = N$ and $r = r_S$ and

$$G > r_S pD.$$

So the ask is driven by learning/risk appetite rather than romance.

**From types to observable probabilities**

Chris does not observe $t$ or $r$, only Dumpling's action $A$. Define the indicator function

$$\mathbf{1}[\text{statement}] = \begin{cases} 1 & \text{if the statement is true} \\ 0 & \text{if the statement is false.} \end{cases}$$

Then, averaging over the unobserved risk type:

$$\Pr(A = 1 \mid L) = \rho\, \mathbf{1}[B + G > r_A pD] + (1 - \rho)\, \mathbf{1}[B + G > r_S pD],$$
$$\Pr(A = 1 \mid N) = \rho\, \mathbf{1}[G > r_A pD] + (1 - \rho)\, \mathbf{1}[G > r_S pD].$$

These expressions say: the probability of asking is a weighted average of whether each risk type would ask.

**Belief updating (what Chris infers from asking)**

After observing $A = 1$, Chris updates using Bayes' Rule:

$$\Pr(L \mid A = 1) = \frac{\Pr(A = 1 \mid L)\,\pi}{\Pr(A = 1 \mid L)\,\pi + \Pr(A = 1 \mid N)\,(1 - \pi)}.$$

Asking is a strong romance signal when $\Pr(A = 1 \mid L)$ is large and $\Pr(A = 1 \mid N)$ is small. When risk-seeking non-romantics ask frequently (large $\Pr(A = 1 \mid N)$), the signal becomes noisy.

**Parameter diagram (the noisy-signal regions)**

The diagram below uses the $(G, B)$ plane (learning benefit vs. bonding benefit), holding $(p, D, r_A, r_S)$ fixed.

Romantic asking thresholds are

$$B = r_S pD - G \quad \text{and} \quad B = r_A pD - G,$$

and non-romantic asking thresholds are

$$G = r_S pD \quad \text{and} \quad G = r_A pD.$$

**Bottom line**

In this model, "she asked for a skate lesson" combines two forces: romantic value ($B$) and learning/risk appetite ($G$ and $r$). Because a risk-averse romantic can choose $A = 0$ and a risk-seeking non-romantic can choose $A = 1$, the action is informative but not definitive: it is a noisy signal of romantic interest.

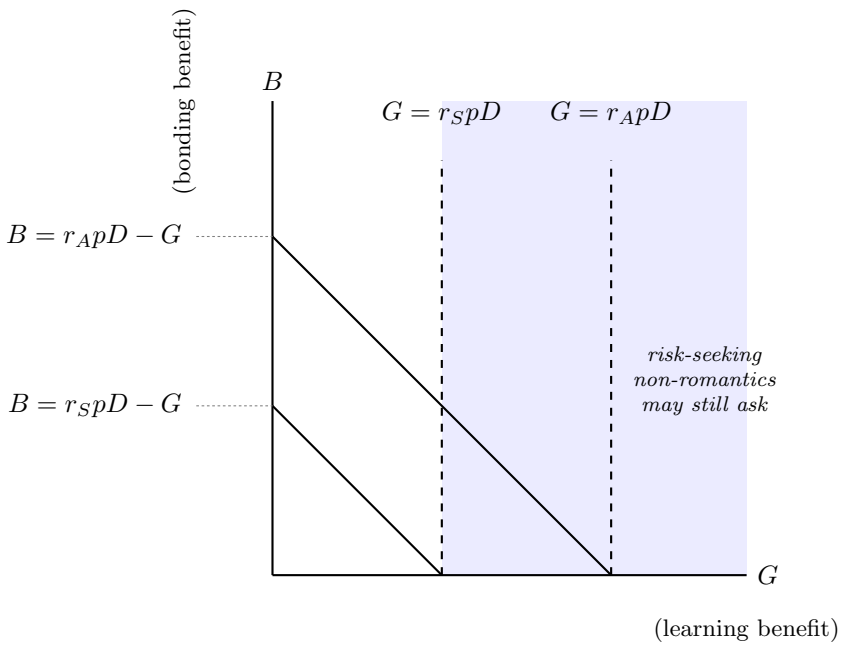



Figure 1: Solid diagonal lines are the *romantic* asking thresholds ($B + G = rpD$); a romantic type asks whenever she sits above the line for her own risk type $r$. Dashed vertical lines are the *non-romantic* asking thresholds ($G = rpD$); a non-romantic type asks whenever she sits to the right of her own line. Because $r_A > r_S$, the risk-seeking ($r_S$) lines sit closer to the origin than the risk-averse ($r_A$) lines — risk-seeking types need a smaller benefit to justify asking. The shaded band ($G > r_S pD$) is the *noisy-signal region*: any point there is asked from by a risk-seeking non-romantic even though $B$ may be zero, which is exactly what makes the action $A = 1$ an imperfect signal of romantic interest.